

DEPI CAPSTONE PROJECT 2025

Egypt Circular Economy

FROM TRASH TO TREASURE

Egypt's Journey Toward a Circular Economy

Turning Waste Data into Action — A Story of Recycling, Challenges & Opportunities

100M+	20.8%	27	60%
Tons Waste/Year	Recycling Rate	Governorates Tracked	2030 Target

Authors: Mohammed Ali | Sondos Yasser | Zeyad Abdelrahman | Khaled Amr | Mennatallah Khaled

Program: Digital Egypt Pioneers Initiative (DEPI) — Data Analytics Track

Supervisor: Eng. Mahmoud Seraj | Computik Learning Solutions (CLS)

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EXECUTIVE SUMMARY

Egypt stands at a critical environmental crossroads. With over 100 million tons of solid waste generated annually — enough to fill the Great Pyramid 10 times every year — and a recycling rate of only 20.8%, the gap between where Egypt is and where it needs to be is enormous. The national 2030 target of 60% recycling represents a near-tripling of current performance.

This booklet documents a comprehensive, data-driven analytical project developed under the Digital Egypt Pioneers Initiative (DEPI) Capstone Program. Using Power BI dashboards, advanced DAX measures, star schema data modeling, and Prophet forecasting models, our team analyzed waste generation, composition, recycling performance, and infrastructure gaps across all 27 Egyptian governorates over a 5-year period.

The findings are urgent — but also deeply optimistic. The data reveals not only where Egypt is failing, but precisely how, where, and why — and points to five concrete, evidence-based solutions that together could transform Egypt's waste economy.

"By 2030, let's not say 'we should have.' Let's say 'we did.' — Turn Egypt's waste into wealth, starting today."

5 Dashboards Live Analytical Tools	5 Years Data Coverage	27 Gov. Full National Scope	5 Solutions Ready to Deploy
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Egypt's Waste Crisis: The Garbage That Could Light a City

Cairo alone generates 3.35 million tons of waste every year. If converted to energy, this single city's waste could power 500,000 homes. Instead, 80% of it ends up in landfills, leaking toxins into the soil and groundwater.

This is not just an environmental problem — it's an economic catastrophe. Egypt is sitting on billions of dollars of recoverable material value, watching it decay in open dumps while paying to manage the mess.



Dashboard Slide 2: Cairo's Hidden Resource — The Garbage That Could Light a City

THE NATIONAL PICTURE AT A GLANCE



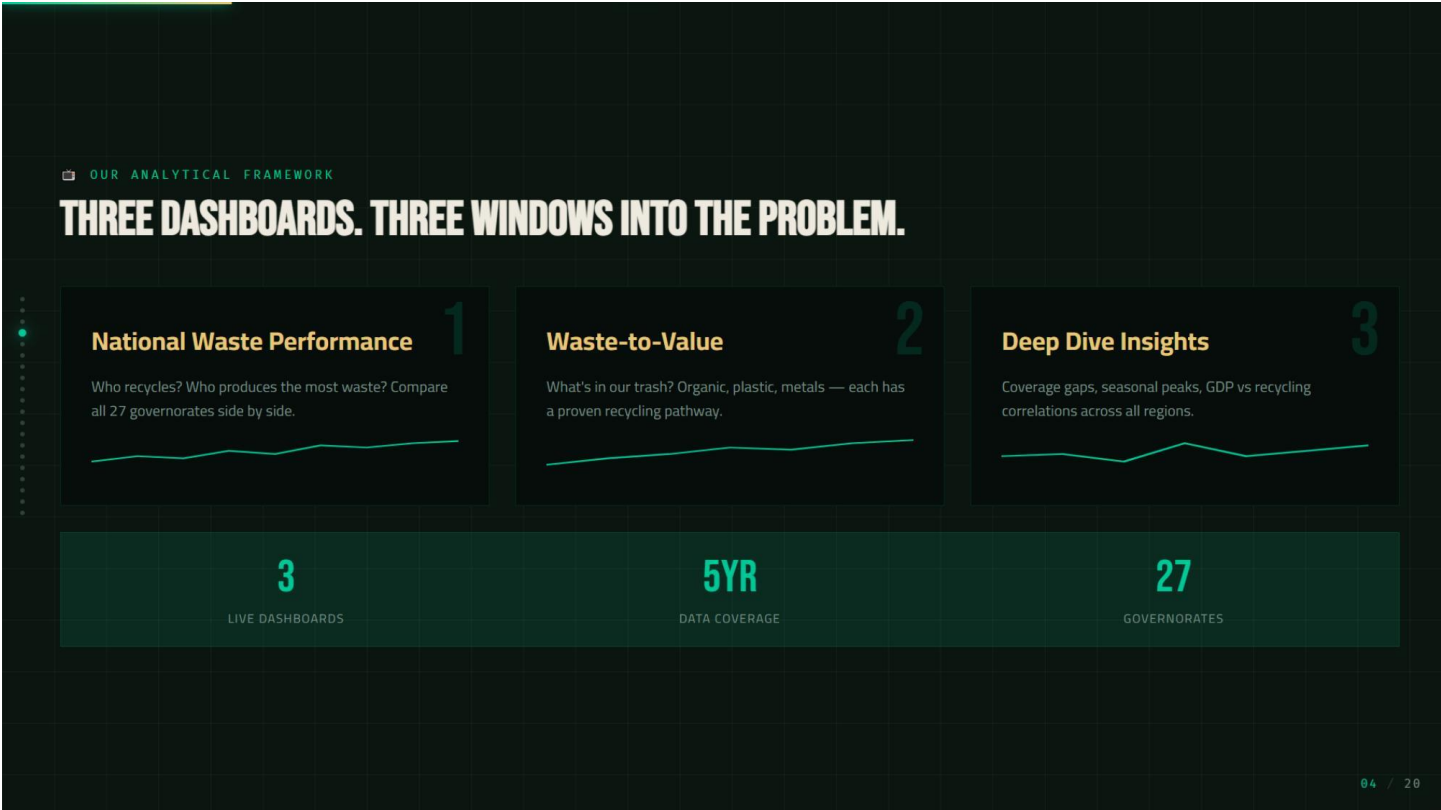
Dashboard Slide 3: Egypt by the Numbers — National Waste Statistics

Metric	Current Value	National Target (2030)	Gap
Total waste generated/yr	100M+ tons	—	—
Current recycling rate	20.8%	60%	39.2 percentage points
Waste to landfill/open dump	80%	<40%	40 percentage points
Waste collection coverage	46.27%	100%	53.73%
Governorates tracked	27	27	Full coverage ✓

"We cannot reach 60% without understanding where we are today."

Three Dashboards. Three Windows Into the Problem.

Our analytical approach was built around three distinct but interconnected Power BI dashboards, each designed to answer a different dimension of the waste management challenge. Together, they form a complete intelligence system covering national performance, material economics, and deep operational insights.



Dashboard Slide 4: Our Analytical Framework Overview

Dashboard	Focus Area	Key Questions Answered
1 — National Waste Performance	Governorate-level comparison	Who recycles? Who produces most? Which regions lag?
2 — Waste-to-Value	Material composition analysis	What's in our trash? What's the economic recovery value?
3 — Deep Dive Insights	Coverage, seasonality, GDP correlation	What drives performance? What are the systemic gaps?

Technical Architecture Summary

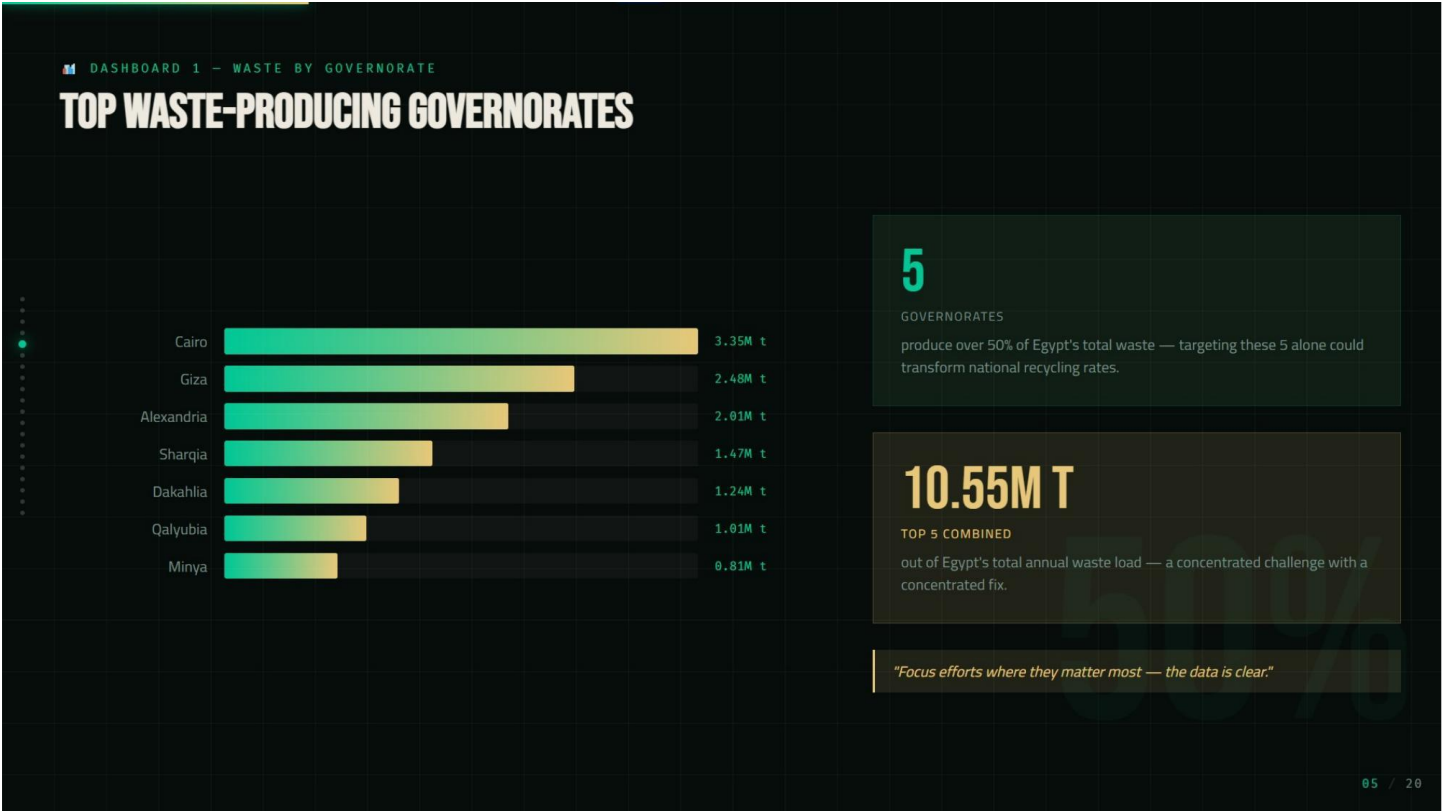
- Data sources: facility records, sorting logs, processing stage records, administrative datasets

- Data preparation: Power Query — cleaning, deduplication, normalization, feature engineering
- Data modeling: Star schema with central ProcessingEvents fact table + 5 dimension tables
- Analytics: Advanced DAX measures including time-intelligence, running totals, ratio measures
- Forecasting: Facebook Prophet model with <2% error rate, 12-month forecast horizon
- Visualization: 3 interactive Power BI dashboard pages with cross-filtering and drill-through

National Waste Performance: Who Recycles? Who Doesn't?

The first dashboard tells the story of Egypt's 27 governorates in numbers — which produce the most waste, which recycle the most, and what the trajectory looks like over time. The patterns that emerge are both alarming and instructive.

TOP WASTE-PRODUCING GOVERNORATES



Dashboard Slide 5: Top Waste-Producing Governorates

Five governorates alone — Cairo, Giza, Alexandria, Sharqia, and Dakahlia — produce over 50% of Egypt's total annual waste. Targeting these five with focused infrastructure investment could transform national recycling rates without requiring a complete nationwide overhaul.

Rank	Governorate	Annual Waste Volume	Strategic Priority
#1	Cairo	3.35M tons	Critical — National capital, highest volume

#2	Giza	2.48M tons	Critical — Adjacent to Cairo, urban density
#3	Alexandria	2.01M tons	High — Major port city, coastal risk
#4	Sharqia	1.47M tons	High — Delta region, agricultural waste mix
#5	Dakahlia	1.24M tons	High — Dense population, limited infrastructure
#6	Qalyubia	1.01M tons	Medium — North Cairo fringe
#7	Minya	0.81M tons	Medium — Upper Egypt hub
TOP 5 COMBINED	—	10.55M tons	= Concentrated fix for a concentrated challenge

"Focus efforts where they matter most — the data is clear."

RECYCLING LEADERS: THE PROOF IT WORKS



Dashboard Slide 6: Top 5 Recycling Governorates (2021–2025)

While the national average sits at ~19% recycling rate (2025), five governorates are already outperforming. These are the 'early adopters' — the proof that improvement is possible with the right systems in place.

Rank	Governorate	Status	Key Learning
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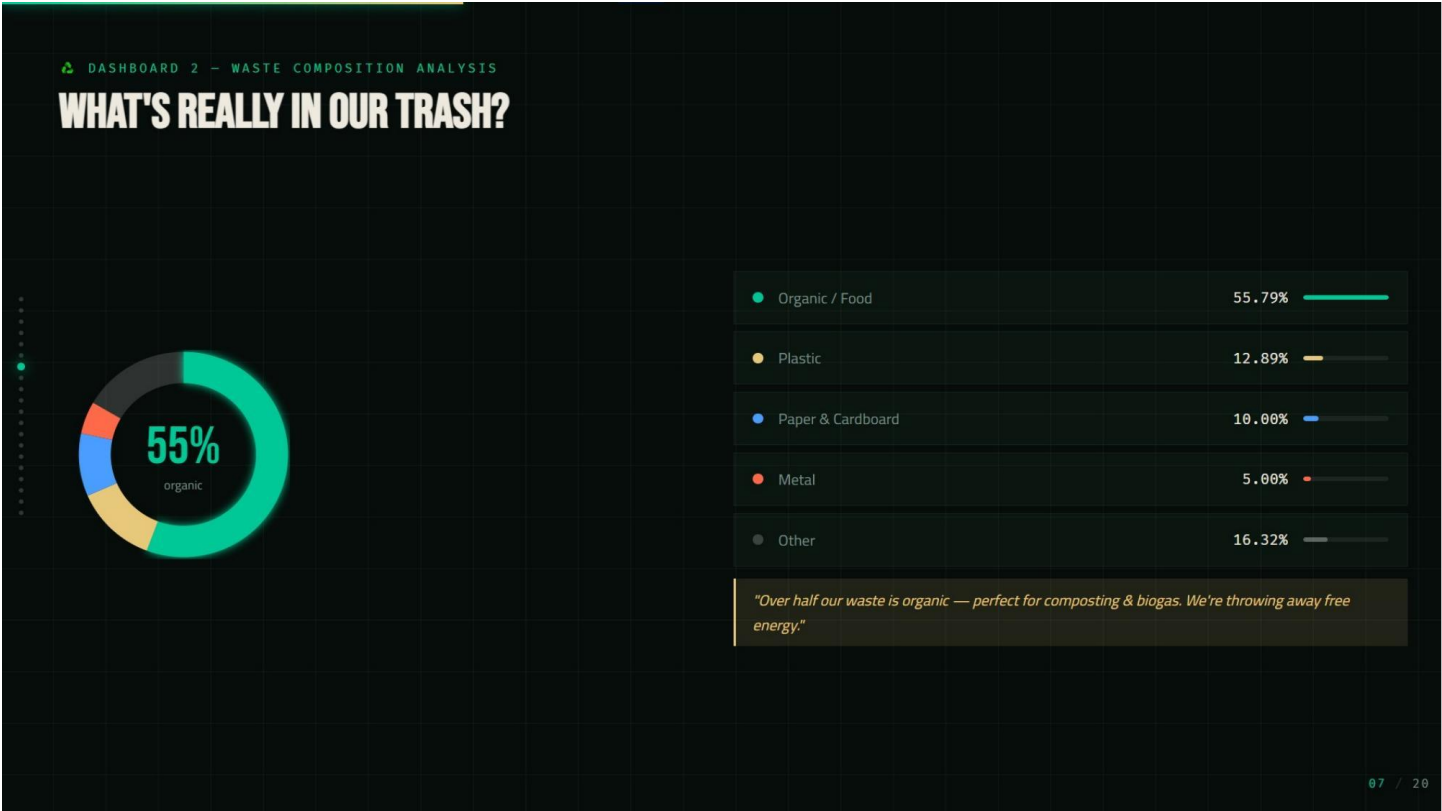
01	Matrouh	Leading nationally	Low population density, strong community programs
02	Sharqia	High performer	Agricultural waste converted to compost
03	North Sinai	Strong growth	Regional waste authority effectiveness
04	Dakahlia	Consistent performer	NGO partnerships in material recovery
05	Sohag	Rising star	Upper Egypt model — replicable in similar regions

National recycling rate grew from approximately 13% in 2021 to approximately 19% in 2025 — meaningful progress, but still far below the 60% target. The gap requires systematic structural change, not incremental improvement.

Waste-to-Value: What's Really in Our Trash?

Understanding what Egypt throws away is the first step to recovering its value. Dashboard 2 breaks down the composition of Egypt's waste stream — and the picture it paints is one of extraordinary, untapped opportunity.

WASTE COMPOSITION BREAKDOWN



Dashboard Slide 7: Waste Composition — What's Really in Our Trash?

Material Type	Share of Waste	Recycling Pathway	Current Status	Opportunity
Organic / Food	55.79%	Composting, biogas	Only 101K/470K tons composted	MASSIVE — 370K ton gap
Plastic	12.89%	Mechanical recycling, RDF	Low recovery rate	HIGH — Growing stream
Paper & Cardboard	10.00%	Paper mills, fiber recovery	Moderate recovery	MEDIUM — Established market

Metal	5.00%	Smelting, metal recovery	Informal sector active	HIGH — High economic value
Other (glass, textiles)	16.32%	Various pathways	Mixed/low recovery	MODERATE

"Over half our waste is organic — perfect for composting and biogas. We're throwing away free energy."

THE ORGANIC WASTE OPPORTUNITY

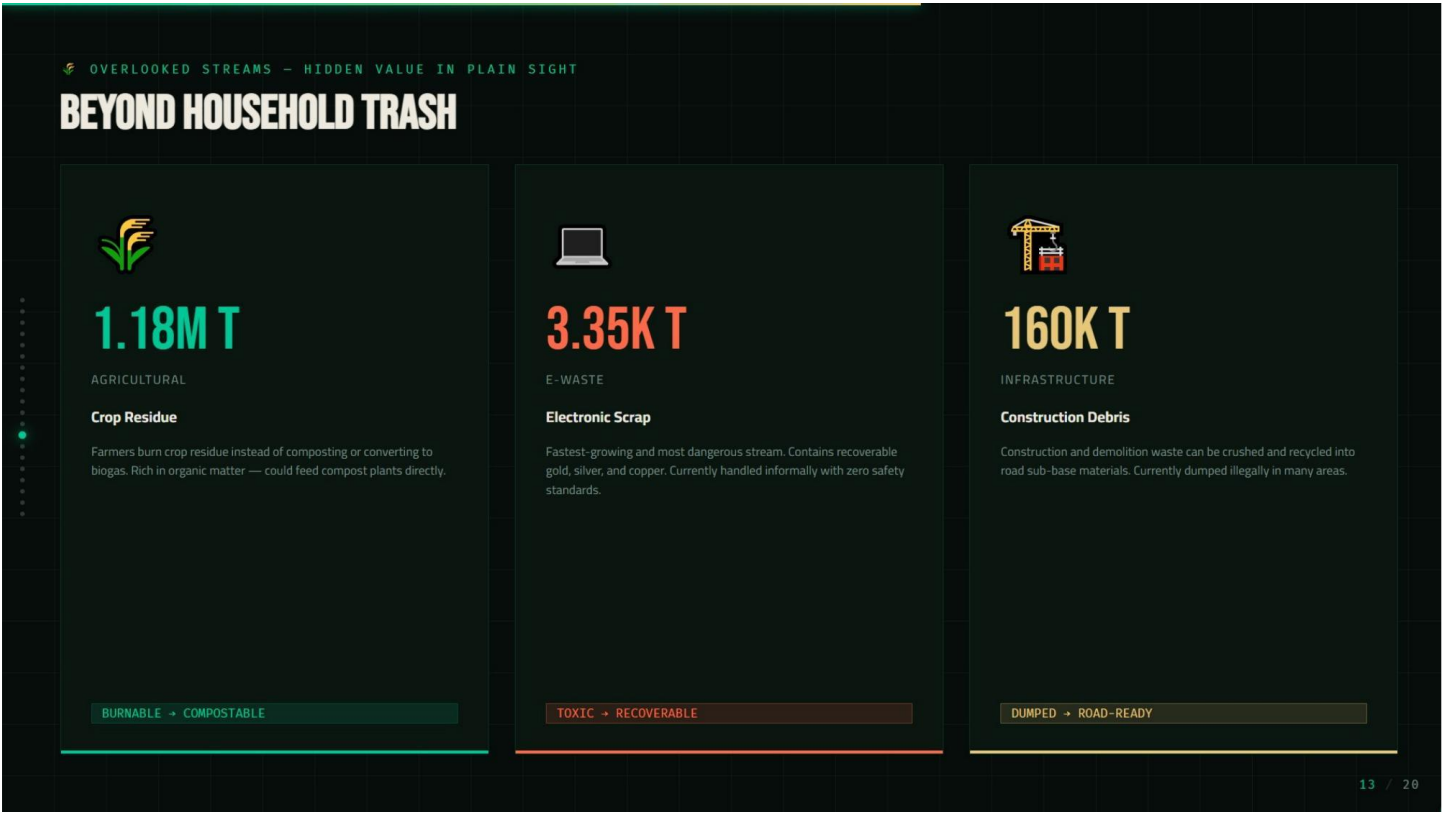


Dashboard Slide 12: 55% Organic — Why We Should Celebrate

470K T Organic Waste Produced	101K T Currently Composted	370K T Composting Gap	0.5T CO₂ Saved per Ton Composted
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Egypt is composting less than one quarter of its organic waste. Every ton of compost that doesn't get produced represents lost fertilizer, lost renewable energy, and 0.5 tons of unnecessary CO₂ emissions. Decentralized composting units in all 27 governorates could close this gap and generate carbon credits for regional revenue.

BEYOND HOUSEHOLD TRASH: OVERLOOKED WASTE STREAMS



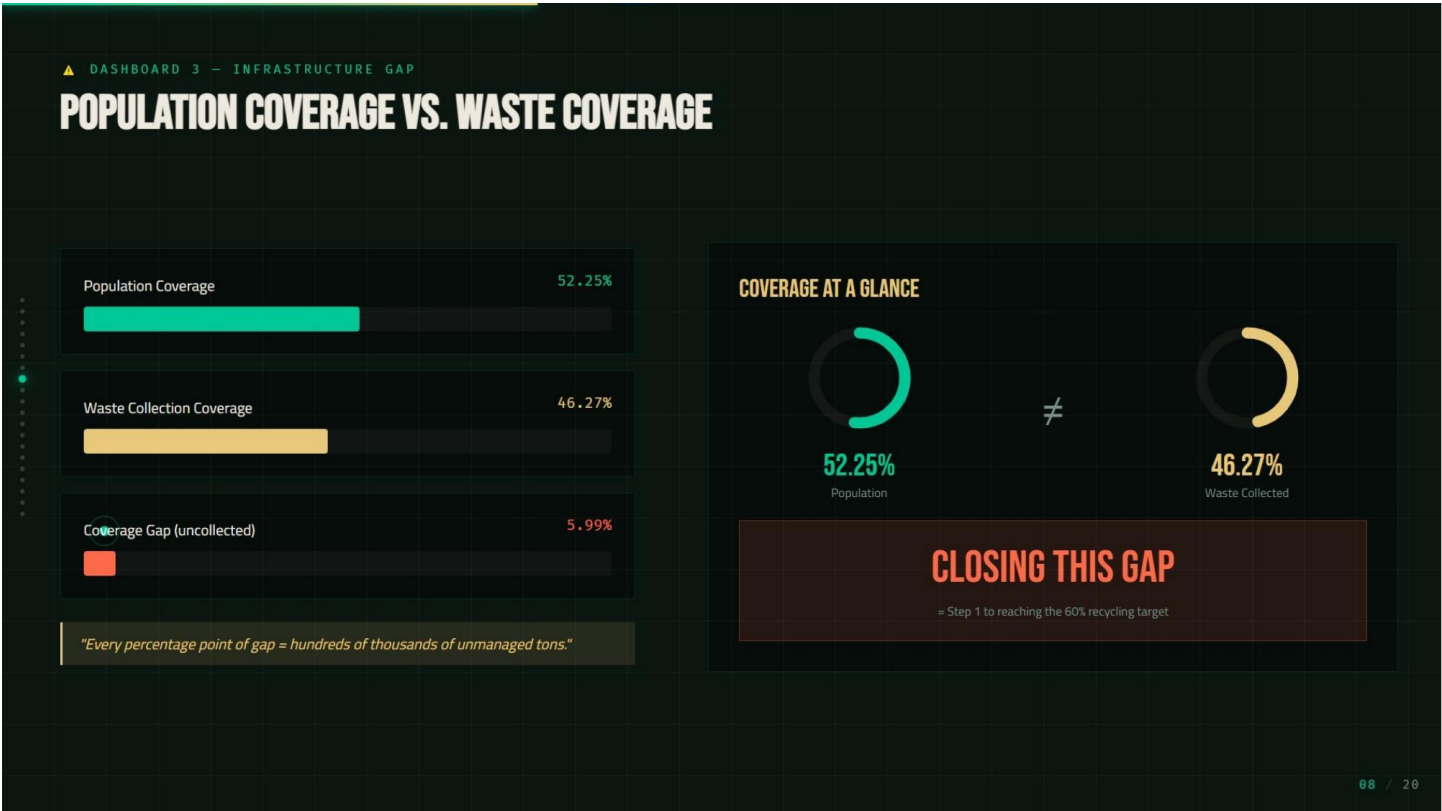
Dashboard Slide 13: Beyond Household Trash — Hidden Value Streams

Waste Stream	Volume	Current Status	Transformation Potential
Agricultural (Crop Residue)	1.18M tons	Burned in fields — toxic air pollution	Compostable, biogas feedstock
E-Waste (Electronic Scrap)	3.35K tons	Informal handling, zero safety standards	Gold, silver, copper recovery
Construction Debris	160K tons	Illegally dumped in many areas	Crushed for road sub-base materials

Deep Dive Insights: Patterns That Demand Action

The third dashboard goes beneath the surface of aggregate statistics to reveal the structural patterns that explain why Egypt's recycling performance is where it is — and where targeted intervention will have the highest impact.

INFRASTRUCTURE GAP: POPULATION VS. WASTE COVERAGE



Dashboard Slide 8: Population Coverage vs. Waste Collection Coverage

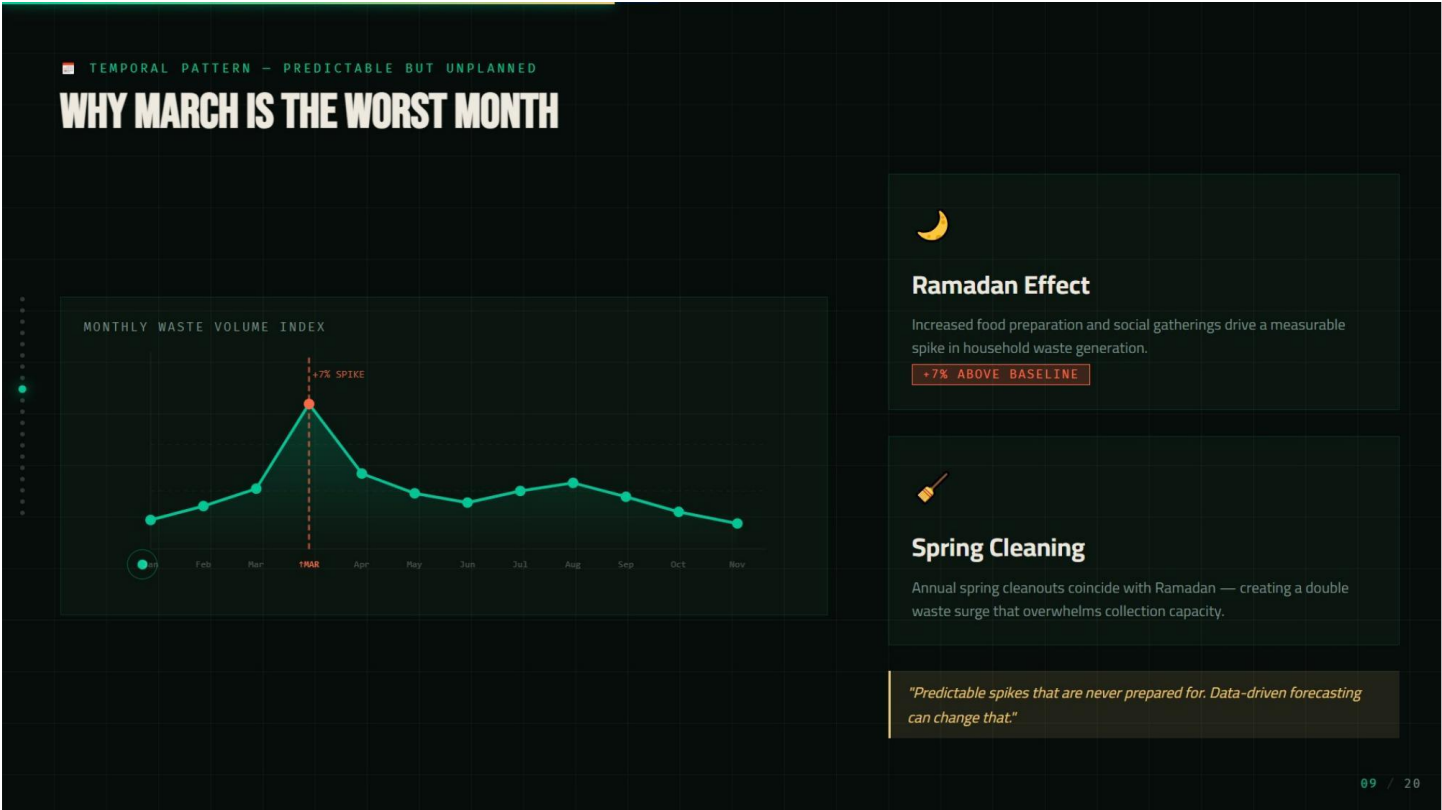
Even at the most basic level of waste collection, Egypt faces a significant structural gap. Only 46.27% of waste is formally collected, despite 52.25% of the population being theoretically covered. This 5.99-percentage-point gap (the 'uncollected gap') represents hundreds of thousands of tons of unmanaged waste annually.

52.25%	46.27%	5.99%	Step 1
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Population Covered	Waste Actually Collected	Coverage Gap (Uncollected)	Closing This Gap = Path to 60%
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"Every percentage point of gap = hundreds of thousands of unmanaged tons."

TEMPORAL PATTERNS: WHY MARCH IS THE WORST MONTH



Dashboard Slide 9: Monthly Waste Volume Index — The Ramadan Effect

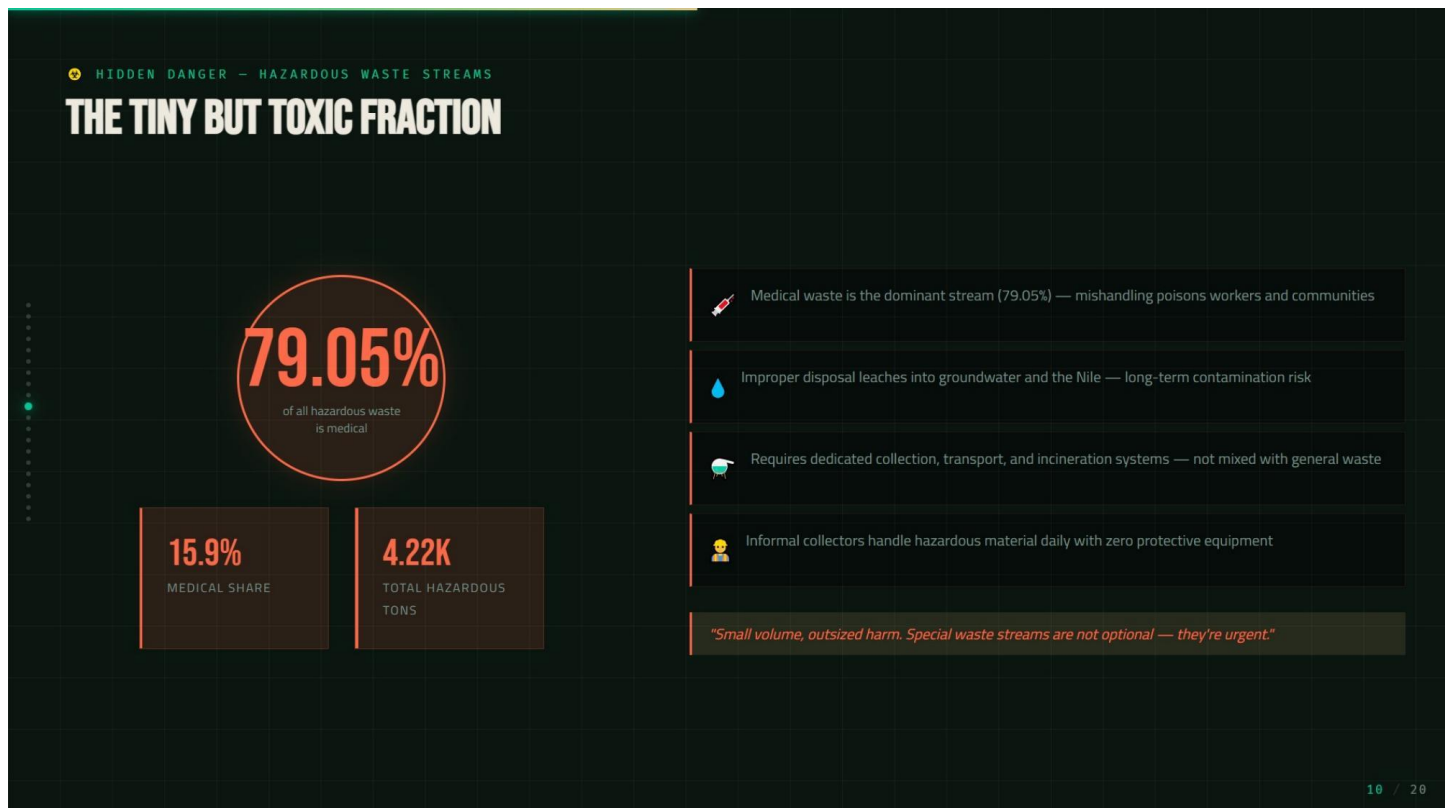
Waste volume in Egypt is not random — it follows highly predictable seasonal patterns that are currently completely unprepared for. The data reveals a consistent +7% spike above baseline in March, driven by two converging factors:

- Ramadan Effect: Increased food preparation, social gatherings, and household activity drive a measurable surge in organic waste generation during Ramadan, which typically falls in March.
- Spring Cleaning: Annual household cleanouts coincide with Ramadan — creating a double waste surge that overwhelms collection capacity.

The tragedy is that these spikes are completely predictable — yet the system consistently fails to prepare for them. Prophet forecasting models can predict these events 12 months ahead with less than 2% error, enabling proactive resource allocation rather than reactive crisis management.

"Predictable spikes that are never prepared for. Data-driven forecasting can change that."

HAZARDOUS WASTE: THE TINY BUT TOXIC FRACTION



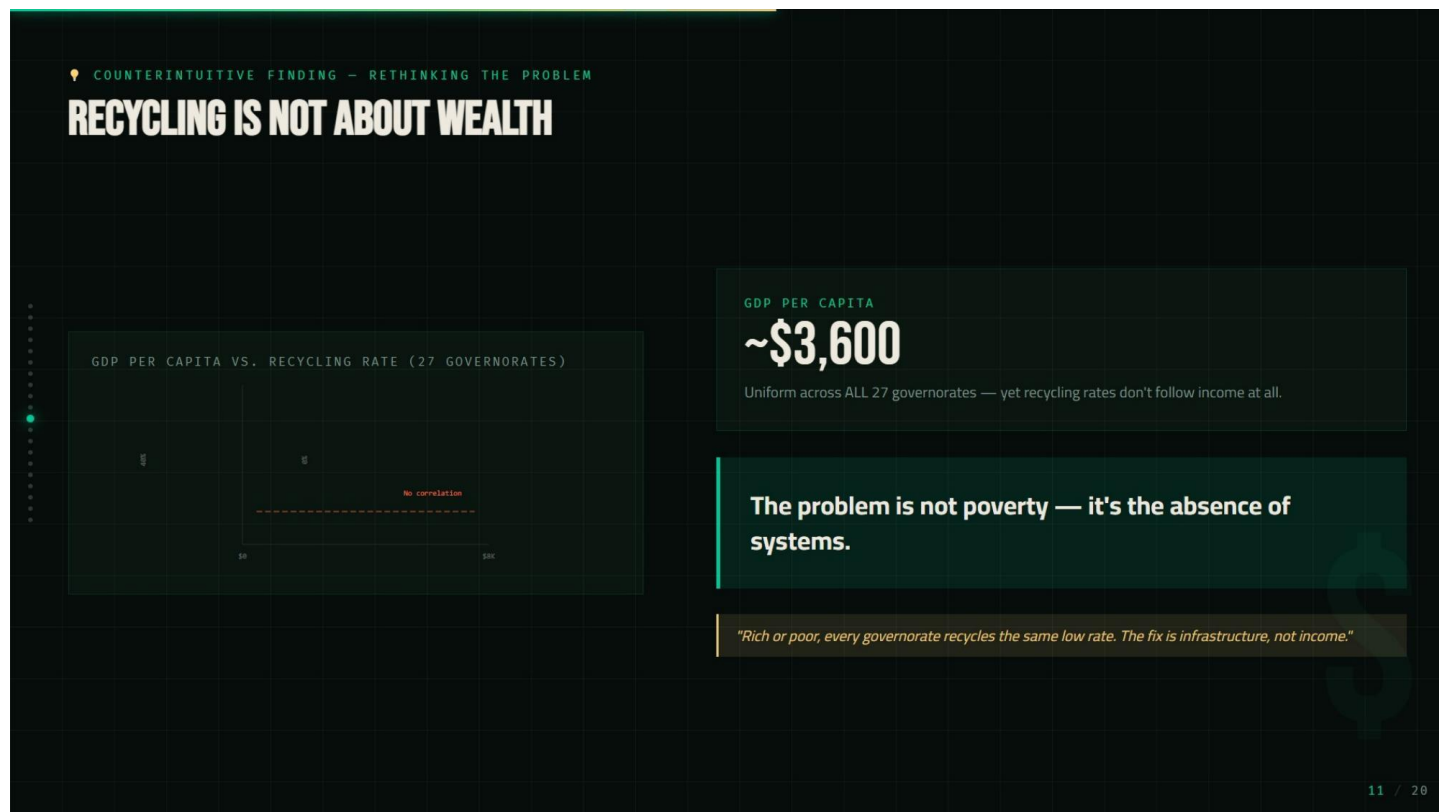
Dashboard Slide 10: The Tiny but Toxic Fraction — Hazardous Waste Streams

While hazardous waste represents a small fraction by volume (4,220 total tons in dataset), its outsized harm demands urgent attention:

- 79.05% of all hazardous waste is medical waste — the dominant and most dangerous stream
- Improper disposal leaches into groundwater and the Nile — long-term contamination risk
- Requires dedicated collection, transport, and incineration systems — cannot mix with general waste
- Informal collectors handle hazardous material daily with zero protective equipment

"Small volume, outsized harm. Special waste streams are not optional — they're urgent."

THE COUNTERINTUITIVE FINDING: RECYCLING IS NOT ABOUT WEALTH



Dashboard Slide 11: GDP Per Capita vs. Recycling Rate — No Correlation

Perhaps the most important finding of the entire analysis: there is no correlation between GDP per capita and recycling rate across Egypt's 27 governorates. GDP per capita is remarkably uniform at approximately \$3,600 across all regions — yet recycling rates vary dramatically.

This finding is deeply significant: it means the recycling gap is NOT a poverty problem. Rich and poor governorates recycle at the same low rate. The problem is the absence of systems — collection infrastructure, sorting facilities, public education, enforcement. This means the solution does not require economic development first — it requires infrastructure and institutional investment, which is within the government's direct control.

The problem is not poverty — it's the absence of systems.

Key Analytical Findings from the Technical Study

The accompanying peer-reviewed research paper ('A Data-Driven Framework for Analyzing Waste Recycling Operations in Egypt') provides additional operational findings from the facility-level analysis. These findings focus on the processing stage of the recycling pipeline — inside facilities rather than at the governorate level.

OPERATIONAL PERFORMANCE FINDINGS

Finding	Metric	Implication
Collection Efficiency	Mean 73%, high variance	Vehicle availability and routing are the bottleneck
Sorting Accuracy Impact	+1% sorting = +2.3% recovery	Front-end quality is the highest-leverage intervention
Top 3 Downtime Causes	67% of all downtime hours	Mechanical failure, feedstock contamination, maintenance overruns
Facility Performance Range	Top sites: 40%+ above portfolio mean	Best-practice transfer can lift all sites without new capex
Seasonal Performance Troughs	Mondays + summer months worst	Staffing patterns and heat affect vehicle performance

"A one-percentage-point improvement in sorting accuracy corresponds to an estimated 2.3% increase in marketable recovered material volume."

DATA QUALITY ASSESSMENT RESULTS

The initial data quality assessment across all ingested data streams identified three major issues that must be addressed for accurate ongoing analytics:

- Completeness gaps: Several facility records had non-null field compliance rates below acceptable thresholds
- Categorical inconsistencies: Material type labels were inconsistently encoded across facilities — requiring standardization
- Temporal misalignment: Timestamps across related records were not synchronized to a consistent UTC format

A data governance framework with standardized field definitions, mandatory validation at data entry, and regular audit cycles is a prerequisite for scaling this analytical framework nationally.

Data Methodology & Technical Architecture

This chapter documents the complete technical pipeline that powers the analytical framework — from raw data ingestion through to the interactive dashboards and forecasting models.

DATA SCHEMA ARCHITECTURE — STAR SCHEMA

The analytical data model uses a star schema: a central fact table surrounded by dimension tables. This architecture enables fast aggregation across multiple analytical dimensions without the complexity of normalized relational schemas.

Entity Type	Table Name	Key Attributes	Purpose
FACT	ProcessingEvents	Volume, Duration, RecoveryRate	Central transaction records of waste processing events
DIMENSION	DimTime	Date, Week, Month, Season	Temporal slicing and time-intelligence calculations
DIMENSION	DimFacility	FacilityID, Location, Capacity	Geographic and facility-level filtering
DIMENSION	DimMaterial	MaterialType, Category, Value	Material composition and economic value analysis
DIMENSION	DimOperator	OperatorID, Shift, ExperienceLevel	Workforce performance tracking

DAX MEASURES DEVELOPED

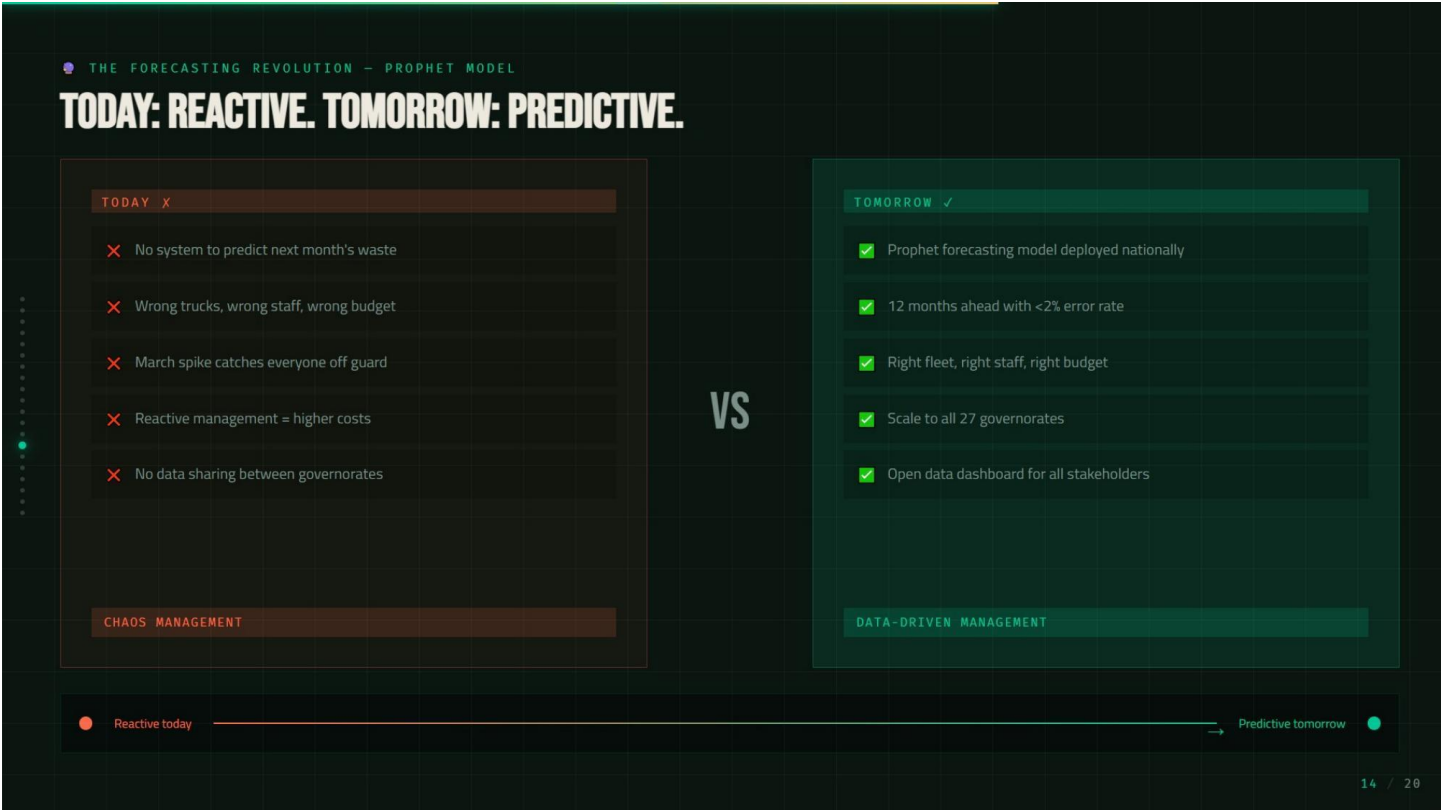
DAX Measure Category	Measures Created	Analytical Purpose
Time Intelligence	Period-over-period comparisons, YTD/MTD/QTD	Trend analysis and seasonal decomposition
Recovery Yield	Cumulative running totals, recovery rates by material	Material value tracking
Ratio Measures	Sorting accuracy %, contamination rates	Quality control KPIs
Forecasting Integration	Prophet model output incorporated as calculated table	Predictive planning

DASHBOARD PAGES ARCHITECTURE

Page	Target Audience	Key Visuals	Cross-filter Capability
Executive Summary	Senior leadership	KPI cards, trend sparklines, geographic heat map	Global — all pages respond
Downtime & Operational Analysis	Operations managers	Pareto chart, drill-through to incident level	Facility, time, cause category
Operator & Resource Performance	Facility supervisors	Scatter plots, matrix visuals, cohort comparisons	Operator, shift, material type

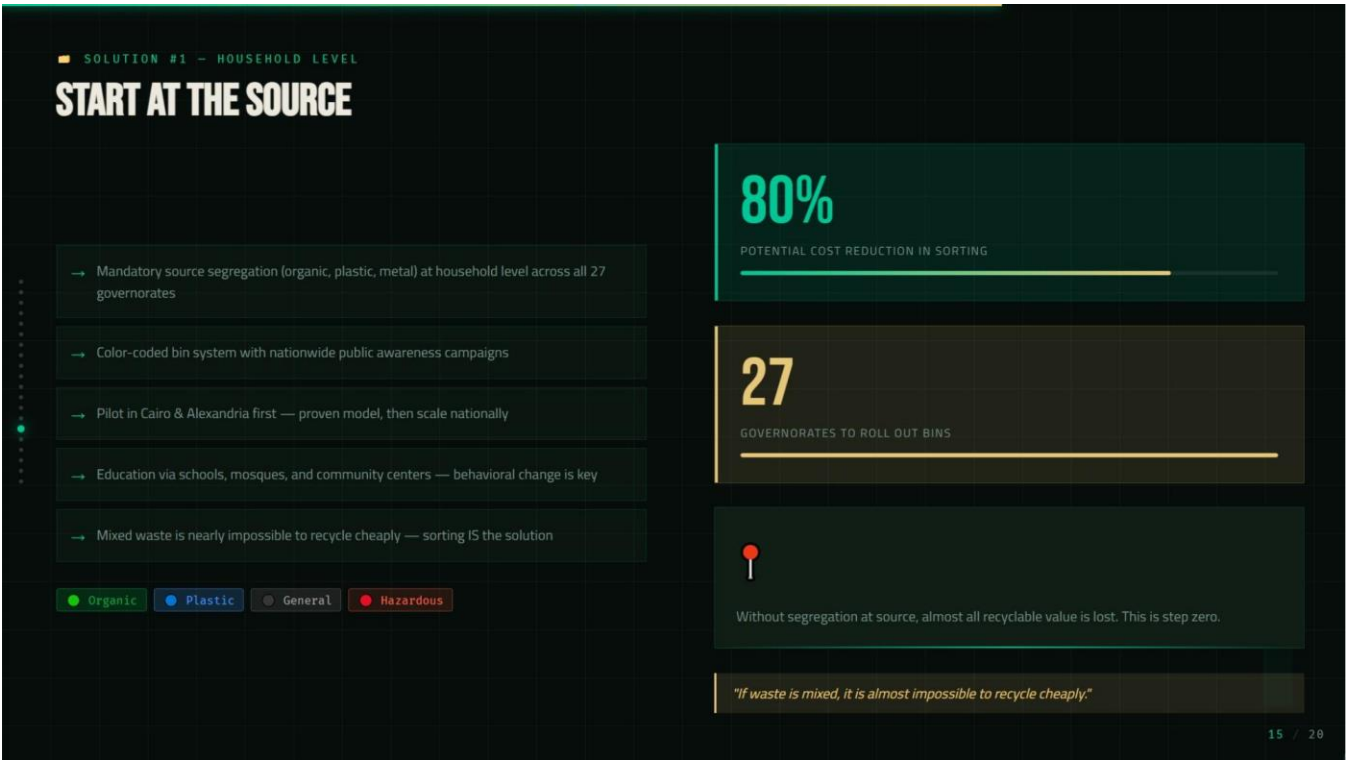
Five Solutions to Transform Egypt's Waste Economy

The data tells us not just where Egypt is failing — it tells us exactly what to do about it. The following five recommendations emerge directly from the analytical findings and together represent a complete roadmap to achieving the 60% recycling target by 2030.



Dashboard Slide 14: Today Reactive vs Tomorrow Predictive — The Transformation Vision

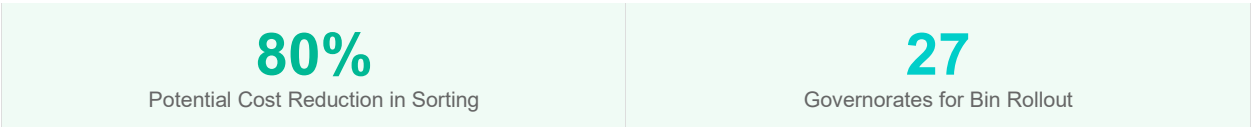
SOLUTION #1 — START AT THE SOURCE: MANDATORY HOUSEHOLD SEGREGATION



Dashboard Slide 15: Start at the Source — Household-Level Segregation

The single most impactful intervention at zero incremental processing cost. Mixed waste is nearly impossible to recycle economically — source segregation is not optional, it is step zero.

- Mandatory source segregation (organic, plastic, metal) at household level across all 27 governorates
- Color-coded bin system with nationwide public awareness campaigns in schools, mosques, and community centers
- Pilot in Cairo and Alexandria first — largest volume, proven model, then scale nationally
- Behavioral change is key: education via schools, mosques, and community centers



SOLUTION #2 — TURN ORGANIC WASTE INTO GOLD: THE COMPOSTING CHAIN



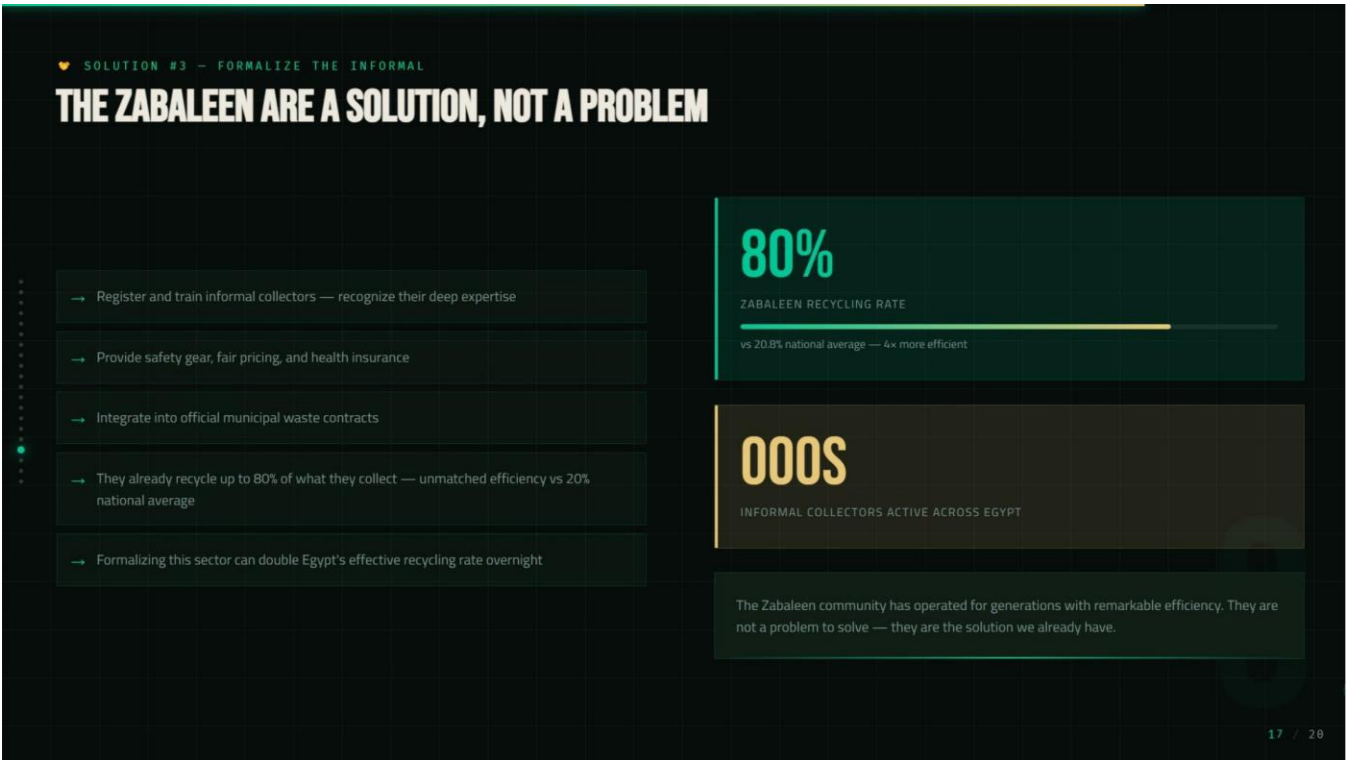
Dashboard Slide 16: Turn Organic Waste into Gold — The Organic Value Chain

With 55% of Egypt's waste being organic, composting and biogas conversion represent the single largest opportunity in the entire waste economy. Currently, Egypt composts less than one quarter of its organic waste, discarding free fertilizer and renewable energy.

- Build decentralized composting units in every governorate — not just large cities
- Convert organic waste to biogas for rural areas with no grid access — energy independence
- Compost subsidies for farmers — replaces expensive imported chemical fertilizers
- Reduces landfill pressure by up to 55% immediately upon deployment
- Each ton of compost generates carbon credits — additional revenue for governorates

55% Landfill Pressure Relieved	0.5T CO₂ Saved per Ton Composted	27 Governorates for Decentralized Units
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SOLUTION #3 — FORMALIZE THE INFORMAL: THE ZABALEEN ARE A SOLUTION, NOT A PROBLEM



Dashboard Slide 17: The Zabaleen Are a Solution, Not a Problem

Egypt's informal waste collectors — the Zabaleen community — already achieve 80% recycling rates on what they collect, versus the 20.8% national average. They are four times more efficient than the formal system. Formalizing this sector could double Egypt's effective recycling rate overnight.

- Register and train informal collectors — recognize their deep expertise built over generations
- Provide safety gear, fair pricing, and health insurance — address the human rights dimension
- Integrate into official municipal waste contracts — give them formal status and stable income
- They already recycle up to 80% of what they collect — leverage this capacity immediately
- Formalizing this sector can double Egypt's effective recycling rate without building a single new facility



SOLUTION #4 — MAKE PRODUCERS PAY: POLICY & ECONOMICS

SOLUTION #4 — POLICY & ECONOMICS

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MAKE PRODUCERS PAY

Extended Producer Responsibility

Producers pay for the end-of-life management of plastic, electronics, and batteries. Internationally proven — EU, Korea, Japan all use this model.

LEGISLATION READY

EPR

Deposit-Return System

Consumer deposits on bottles and cans — returned upon recycling. Proven to boost collection rates by 60%+ in every country that deployed it.

+68% COLLECTION

D-R

Tax Incentives

Tax breaks for manufacturers using recycled content. Makes circular materials cost-competitive with virgin resources — no subsidy needed long-term.

MARKET DRIVEN

TAX

"If companies pay for waste, they'll design less of it. Policy is the most powerful recycling tool."

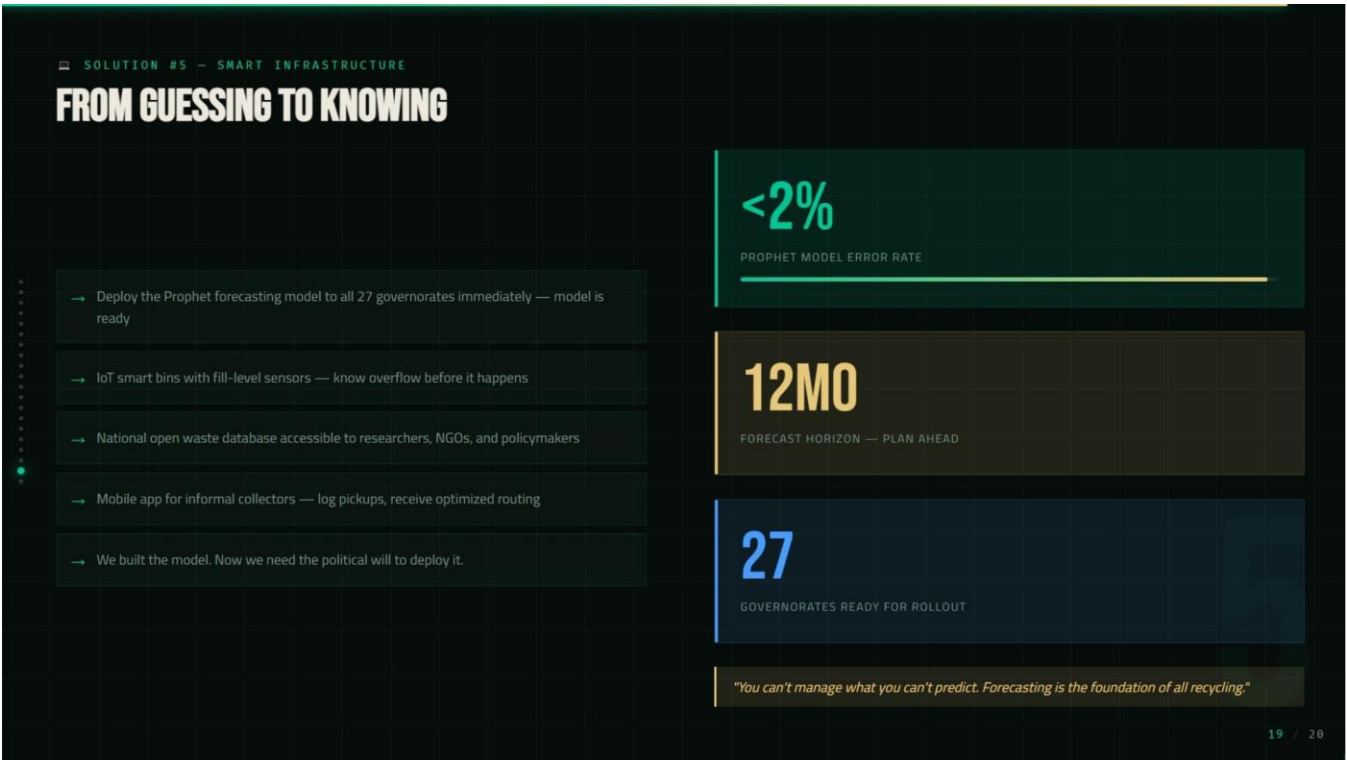
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Dashboard Slide 18: Make Producers Pay — EPR, Deposit-Return, Tax Incentives

- **Extended Producer Responsibility (EPR):** Producers pay for end-of-life management of plastic, electronics, and batteries. Internationally proven — EU, Korea, Japan all use this model. Legislation is ready.
- **Deposit-Return System:** Consumer deposits on bottles and cans — returned upon recycling. Proven to boost collection rates by 60%+ in every country that has deployed it.
- **Tax Incentives:** Tax breaks for manufacturers using recycled content. Makes circular materials cost-competitive with virgin resources — no subsidy needed long-term.

"If companies pay for waste, they'll design less of it. Policy is the most powerful recycling tool."

SOLUTION #5 — FROM GUESSING TO KNOWING: SMART INFRASTRUCTURE & FORECASTING



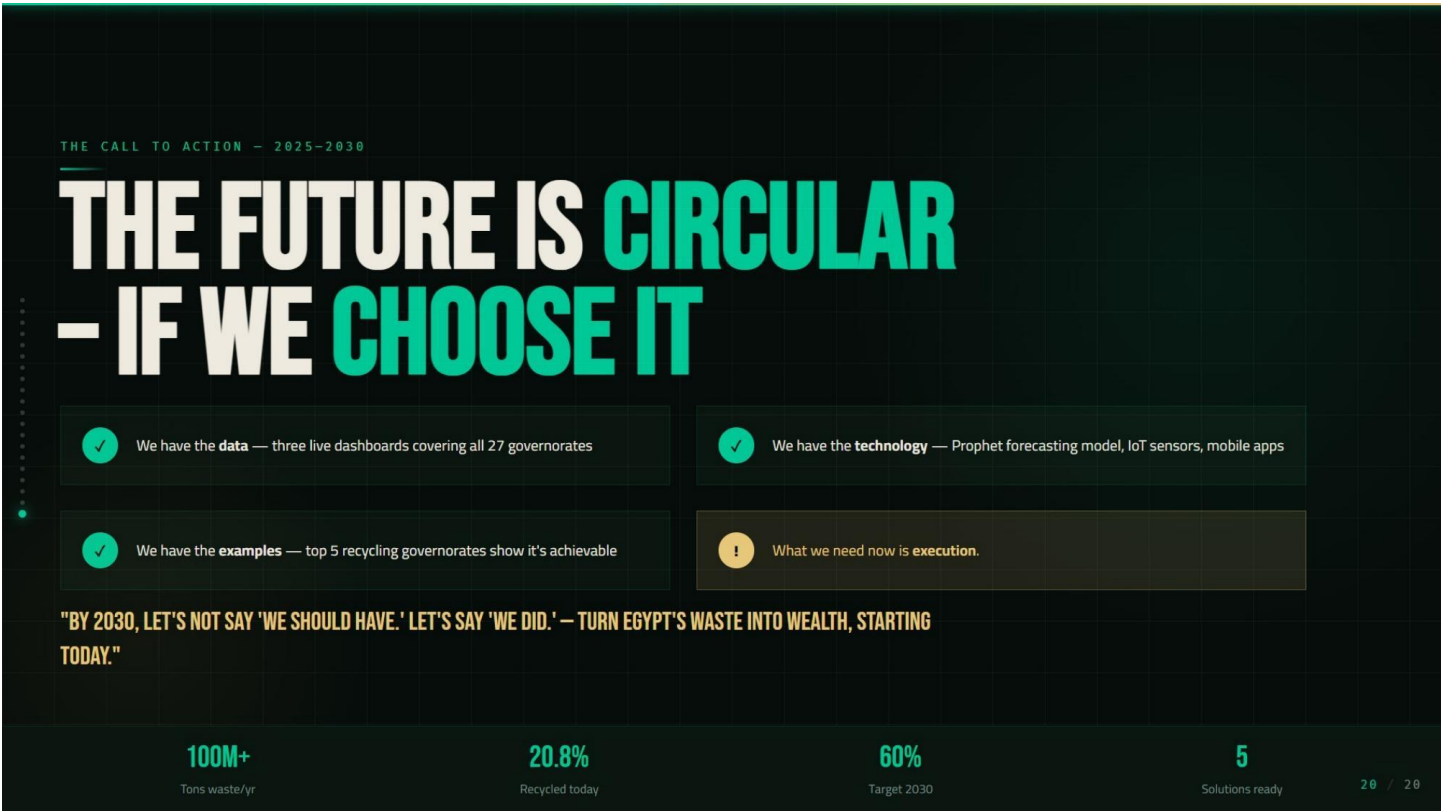
Dashboard Slide 19: From Guessing to Knowing — Smart Infrastructure

Egypt has already built the forecasting model. The Prophet model predicts waste volumes 12 months ahead with less than 2% error. What's missing is the political will to deploy it — and the physical IoT infrastructure to make data-driven management real.

- Deploy the Prophet forecasting model to all 27 governorates immediately — the model is ready
- IoT smart bins with fill-level sensors — know overflow before it happens, optimize routing
- National open waste database accessible to researchers, NGOs, and policymakers
- Mobile app for informal collectors — log pickups, receive optimized routing in real time

<2% Prophet Model Error Rate	12 Months Forecast Horizon	27 Governorates Ready for Rollout
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The Future Is Circular — If We Choose It



Dashboard Slide 20: The Call to Action — 2025–2030

Egypt has everything it needs to achieve the 60% recycling target by 2030. The data is available. The technology exists. The models are built. The informal sector is already demonstrating that 80% recycling is achievable with the right conditions.

We Have the...	Evidence	Next Step
DATA	3 live dashboards covering all 27 governorates, 5 years of trends	Open access, policy integration
TECHNOLOGY	Prophet forecasting (<2% error), IoT sensors ready, mobile apps designed	Political will to deploy
EXAMPLES	Top 5 recycling governorates prove it's achievable at scale	Systematic knowledge transfer
SOLUTIONS	5 ready-to-deploy recommendations with proven international precedents	Funding + execution

"By 2030, let's not say 'we should have.' Let's say 'we did.' — Turn Egypt's waste into wealth, starting today."

ALIGNMENT WITH NATIONAL TARGETS

The cumulative impact of the five recommended interventions is estimated to support a meaningful progression toward the national recycling target of 60% by 2027 (per Egypt Vision 2030). Specifically:

- Improvements in sorting accuracy from source segregation: +8 to +12 percentage points toward recovery rate
- Organic waste composting at scale: +6 to +10 percentage points through reduced landfill diversion
- Zabaleen formalization: +3 to +5 percentage points through increased effective collection
- Predictive management reducing seasonal failures: +2 to +3 percentage points through system reliability

Combined projected improvement: 15 to 20 percentage points above current baseline performance — subject to implementation fidelity and infrastructure readiness. From 20.8% to a projected 36–41% — still below 60%, but a transformation of the baseline from which further progress can accelerate.

ACKNOWLEDGMENTS

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Special recognition is extended to Eng. Mahmoud Seraj for his expert supervision and continuous technical guidance throughout the project lifecycle, and to Computik Learning Solutions (CLS) for providing the structured learning environment and technical resources that enabled this work.

The authors also acknowledge the foundational role of Egypt Vision 2030 in motivating research of this nature and in establishing the national sustainability targets that give this work its strategic significance.

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